

Square-Energy Asymmetry from Packing at Most Two Cycles

Abstract

For a finite simple graph G , let $s^+(G)$ and $s^-(G)$ be the sums of the squares of its positive and negative adjacency eigenvalues. We prove that $s^+(G) > s^-(G)$ whenever G contains a cycle, every cycle of G has length 3 modulo 4, and the maximum size of a collection of pairwise vertex-disjoint cycles is at most two. If every cycle has length 1 modulo 4, the inequality is reversed. The proof groups the Sachs expansion by collections of vertex-disjoint cycles. On the imaginary axis each cycle contributes the phase $-2i^{-\ell}$, while the remaining factor is a signless matching polynomial with positive coefficients. Packing number at most two then puts the normalized characteristic polynomial strictly in one open half-plane. A Coulson identity converts this fixed argument sign into the claimed square-energy inequality. Applications include a strict form of the positive square-energy conjecture for bicyclic block graphs and broad families of triangular cacti.

1 Introduction

Let G be a finite simple graph with adjacency eigenvalues $\lambda_1, \dots, \lambda_n$. Its positive and negative square energies are

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2.$$

Since $\text{tr } A(G)^2 = 2|E(G)|$,

$$s^+(G) + s^-(G) = 2|E(G)|. \tag{1}$$

These quantities were introduced in connection with spectral bounds for the chromatic number by Elphick, Farber, Goldberg, and Wocjan [?]; subsequent structural and asymmetry results appear in [?, ?, ?]. The conjecture that every connected n -vertex graph satisfies $\min\{s^+, s^-\} \geq n - 1$ has recently been proved by Liu, Tang, and Zhang [?]. Akbari, Kumar, Mohar, Pragada, and Zhang proposed the stronger conjecture that a connected graph with at least $n + 1$ edges satisfies $s^+(G) \geq n$ [?, Conjecture 1.2].

For a graph G , write $\nu_o(G)$ for the maximum cardinality of a collection of pairwise vertex-disjoint cycles. Our main result is the following sign theorem. Connectivity is not required.

Theorem 1.1. *Let G be a finite simple graph that contains a cycle and satisfies $\nu_o(G) \leq 2$.*

- (i) *If every cycle of G has length congruent to 3 (mod 4), then $s^+(G) > s^-(G)$.*
- (ii) *If every cycle of G has length congruent to 1 (mod 4), then $s^+(G) < s^-(G)$.*

Ning and Zeng proved the corresponding statement for an odd unicyclic graph [?]. Their argument also uses a matching polynomial and a Coulson-type identity. Their result is the one-cycle case of the sign assertion above; the new issue here is that two-cycle terms can make the real part of the normalized characteristic polynomial negative. They remain real, however, so the

open-half-plane argument survives. Here the full Sachs expansion is grouped by vertex-disjoint cycle collections; the terms with two cycles are real, whereas the terms with one cycle have a common nonzero imaginary sign. This is exactly where the hypothesis $\nu_o(G) \leq 2$ enters. The reverse assertion in Theorem ??(ii) is included as the phase-reversed companion statement. We do not make a novelty or priority claim on the basis of a negative literature search.

Section ?? gives a self-contained proof, including the required Coulson identity and a careful continuous-argument step. Section ?? records the cactus structure forced by the cycle congruence, and Section ?? gives applications. We also use the induced-subgraph superadditivity theorem of Akbari, Kumar, Mohar, and Pragada [?]: if $H_1, \dots, H_r$ are pairwise disjoint induced subgraphs of G , then

$$s^+(G) \geq \sum_{j=1}^r s^+(H_j). \quad (2)$$

2 Sachs expansion on the imaginary axis

For a graph H of order h , let $m_j(H)$ denote its number of j -edge matchings and define the following normalization of the matching polynomial; see Godsil and Gutman [?]:

$$Z_H(t) = \sum_{j=0}^{\lfloor h/2 \rfloor} m_j(H) t^{h-2j}. \quad (3)$$

We use $Z_\emptyset(t) = 1$. Thus $Z_H(t) > 0$ for every $t > 0$. Equivalently, if

$$\mu_H(x) = \sum_j (-1)^j m_j(H) x^{h-2j}$$

is the matching polynomial, then

$$i^{-h} \mu_H(it) = Z_H(t). \quad (4)$$

Let $\mathfrak{C}(G)$ be the set of all collections $\mathcal{C}$ of pairwise vertex-disjoint cycles of G , including the empty collection. Put $V(\mathcal{C}) = \bigcup_{C \in \mathcal{C}} V(C)$.

Lemma 2.1 (Sachs expansion grouped by cycles). *If G has order n and characteristic polynomial $\phi_G(x) = \det(xI - A(G))$, then*

$$\phi_G(x) = \sum_{\mathcal{C} \in \mathfrak{C}(G)} (-2)^{|\mathcal{C}|} \mu_{G-V(\mathcal{C})}(x). \quad (5)$$

Consequently, for $t > 0$,

$$\Psi_G(t) := i^{-n} \phi_G(it) = \sum_{\mathcal{C} \in \mathfrak{C}(G)} \left(\prod_{C \in \mathcal{C}} q_{|C|} \right) Z_{G-V(\mathcal{C})}(t), \quad q_\ell = -2i^{-\ell}. \quad (6)$$

Proof. Sachs' formula [?] sums over elementary subgraphs S , whose components are edges and cycles:

$$\phi_G(x) = \sum_S (-1)^{c(S)} 2^{c_o(S)} x^{n-|V(S)|},$$

where $c(S)$ and $c_o(S)$ are the numbers of components and cycle components of S . Fix its cycle components as a collection $\mathcal{C}$. The other components form a matching of some size j in $G -$

$V(\mathcal{C})$. The corresponding sign and weight are $(-1)^{|\mathcal{C}|+j2^{|\mathcal{C}|}}$. Summing over those matchings gives $(-2)^{|\mathcal{C}|}\mu_{G-V(\mathcal{C})}(x)$, proving (??).

For a fixed $\mathcal{C}$, set $L = \sum_{C \in \mathcal{C}} |C|$. Substitution of it and (??) give

$$i^{-n}(-2)^{|\mathcal{C}|}\mu_{G-V(\mathcal{C})}(it) = (-2)^{|\mathcal{C}|}i^{-L}Z_{G-V(\mathcal{C})}(t),$$

which factors into one $q_{|C|}$ for each $C \in \mathcal{C}$. □

The next identity is a signed square-energy version of the classical Coulson integral formula. We include the derivation to fix both the sign and the branch convention; compare [?, ?].

Lemma 2.2 (Coulson identity). *For $t > 0$, define*

$$\Theta_G(t) = \sum_{j=1}^n \arctan\left(\frac{\lambda_j}{t}\right), \quad -\frac{\pi}{2} < \arctan x < \frac{\pi}{2}. \quad (7)$$

Then the integral below is absolutely convergent and

$$s^+(G) - s^-(G) = -\frac{4}{\pi} \int_0^\infty t \Theta_G(t) dt. \quad (8)$$

Proof. For every real λ ,

$$\lambda|\lambda| = \frac{2}{\pi} \int_0^\infty \frac{\lambda^3}{\lambda^2 + t^2} dt.$$

The scalar integral is absolutely convergent. Since there are finitely many eigenvalues and $\sum_j \lambda_j = \text{tr } A(G) = 0$,

$$s^+ - s^- = \frac{2}{\pi} \int_0^\infty \sum_j \frac{\lambda_j^3}{\lambda_j^2 + t^2} dt = \frac{2}{\pi} \int_0^\infty t^2 \Theta'_G(t) dt.$$

Indeed, $\Theta'_G(t) = -\sum_j \lambda_j/(\lambda_j^2 + t^2)$ and $\sum_j \lambda_j^3/(\lambda_j^2 + t^2) = -t^2 \sum_j \lambda_j/(\lambda_j^2 + t^2)$.

Integrate by parts first on $[\varepsilon, R]$. At zero, $|\Theta_G(t)| \leq n\pi/2$, so $t^2\Theta_G(t) \rightarrow 0$. At infinity, Taylor expansion and $\sum_j \lambda_j = 0$ give $\Theta_G(t) = O(t^{-3})$, so again $t^2\Theta_G(t) \rightarrow 0$. The same estimates show that $t\Theta_G(t)$ is absolutely integrable: it is $O(t)$ at zero and $O(t^{-2})$ at infinity. Letting $\varepsilon \downarrow 0$ and $R \rightarrow \infty$ yields (??). □

Proof of Theorem ??. Suppose first that every cycle length is 3 (mod 4). For such a length,

$$q_\ell = -2i^{-\ell} = -2i.$$

Because $\nu_o(G) \leq 2$, expansion (??) contains only collections of zero, one, or two cycles. The zero-cycle term is real; each two-cycle term is real because $(-2i)^2 = -4$; and every one-cycle term has imaginary part $-2Z_{G-V(C)}(t) < 0$. Since G contains a cycle, it follows that

$$\text{Im } \Psi_G(t) = -2 \sum_{C \text{ a cycle of } G} Z_{G-V(C)}(t) < 0 \quad (t > 0). \quad (9)$$

We now justify the argument without an implicit branch choice. Directly from the eigenvalue factorization,

$$\Psi_G(t) = i^{-n} \prod_j (it - \lambda_j) = \prod_j (t + i\lambda_j). \quad (10)$$

Thus $\Theta_G(t)$ is a continuous argument of $\Psi_G(t)$. By (??), the curve $\Psi_G((0, \infty))$ lies strictly in the open lower half-plane. Its principal argument $\alpha(t) = \text{Arg } \Psi_G(t) \in (-\pi, 0)$ is therefore continuous. Moreover, $\Theta_G(t) \rightarrow 0$ as $t \rightarrow \infty$, while $\Psi_G(t) = t^n(1 + o(1))$ and (??) imply $\alpha(t) \rightarrow 0$ from below. The continuous function $\Theta_G(t) - \alpha(t)$ takes values in $2\pi\mathbb{Z}$, hence is constant; its limit at infinity is zero. Therefore $\Theta_G(t) = \alpha(t) < 0$ for every $t > 0$. In particular the integral in (??) is strictly negative, and $s^+(G) - s^-(G) > 0$.

If every cycle length is $1 \pmod{4}$, then $q_\ell = 2i$. The two-cycle terms are again real, while

$$\text{Im } \Psi_G(t) = 2 \sum_{C \text{ a cycle of } G} Z_{G-V(C)}(t) > 0.$$

The same argument in the open upper half-plane shows that $\Theta_G(t) \in (0, \pi)$ for every $t > 0$. Lemma ?? now gives $s^+(G) - s^-(G) < 0$. $\square$

Remark 2.3. The proof applies verbatim to disconnected graphs. Equivalently, ϕ_G and Ψ_G factor over components and the square energies are additive. Forest components contribute zero asymmetry, while the common cycle phase controls all cyclic components under the global packing hypothesis.

3 The structure forced by one cycle phase

We recall that a cactus is a graph in which every block is either a single edge or a cycle. Isolated vertices are allowed here.

Lemma 3.1. *If every cycle of a finite simple graph G has length congruent to $3 \pmod{4}$, then every cyclic block of G is one of those cycles. In particular, G is a cactus. The same conclusion holds with $3 \pmod{4}$ replaced by $1 \pmod{4}$.*

Proof. A theta graph consists of three internally vertex-disjoint paths between the same two distinct vertices. If their lengths are a, b, c , its three cycles have lengths $a + b$, $a + c$, and $b + c$. These three lengths cannot all be congruent to the same odd residue $r \pmod{4}$, since

$$2a = (a + b) + (a + c) - (b + c) \equiv r \pmod{4},$$

whose left side is even and right side is odd.

It remains to recall why a block that is neither an edge nor a cycle contains a theta subdivision. Let B be such a block. It is 2-connected, and it contains a cycle C . If B has an edge not on C , take a component K of $B - V(C)$ incident with C , or a chord of C if there is no such component. In the former case K has at least two distinct attachment vertices u, v on C : a unique attachment would be a cut vertex of the 2-connected graph B . Since K is connected, a path in K between neighbors of two such attachments, together with its two attachment edges, gives a $u-v$ path internally disjoint from C . In the chord case the chord is such a path. Together with the two $u-v$ arcs of C , this is a theta subdivision, a contradiction. Hence every nontrivial block is an edge or a cycle. $\square$

The lemma also avoids any need for an unproved classification of how triangles can intersect: under the cycle-congruence hypothesis the full block structure follows from the elementary theta obstruction.

4 Applications

For a graph G with $\kappa(G)$ components, its cyclomatic number is

$$\beta(G) = |E(G)| - |V(G)| + \kappa(G).$$

A block graph is a graph every block of which is a clique.

Corollary 4.1 (Bicyclic block graphs). *Let G be a connected block graph with n vertices and cyclomatic number 2. Then G has exactly two blocks isomorphic to K_3 , all its other blocks are copies of K_2 , and*

$$s^+(G) > s^-(G), \quad s^+(G) > |E(G)| = n + 1 > s^-(G).$$

In particular, Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang [?] holds strictly, and in the stronger form $s^+(G) > n + 1$, for every bicyclic block graph.

Proof. Cyclomatic number is additive over the blocks of a connected graph. A clique block K_r contributes

$$|E(K_r)| - |V(K_r)| + 1 = \binom{r}{2} - r + 1 = \frac{(r-1)(r-2)}{2}.$$

This contribution is 0 for $r = 2$, 1 for $r = 3$, and at least 3 for $r \geq 4$. Since the total is 2, there are exactly two K_3 blocks and all other blocks are K_2 's. Thus the only cycles are the two triangles, so their vertex-disjoint packing number is at most two. Theorem ??(i) gives $s^+ > s^-$. Finally $|E(G)| = n - 1 + \beta(G) = n + 1$, and (??) says that the average of s^+ and s^- is $|E(G)|$. The displayed strict inequalities follow. $\square$

The next application goes one cycle rank farther. Its packing-three case is handled by cutting bridges rather than by extending the half-plane argument: three-cycle Sachs terms have the opposite imaginary sign.

Proposition 4.2 (Tricyclic block graphs). *Let G be a connected block graph with n vertices and cyclomatic number 3. Then*

$$s^+(G) > n.$$

Thus Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang holds strictly for every tricyclic block graph.

Proof. As in the proof of Corollary ??, the positive contributions of clique blocks to the cyclomatic number are 1, 3, 6, ... Hence either G has one K_4 block and all other blocks are bridges, or it has three K_3 blocks and all other blocks are bridges.

Suppose first that the unique cyclic block is K_4 . Every cycle lies in this block, and no two cycles are vertex-disjoint. In the normalized Sachs expansion, a triangle has phase $q_3 = -2i$, while a 4-cycle has real phase $q_4 = -2$. Therefore

$$\operatorname{Im} \Psi_G(t) = -2 \sum_{C \subseteq K_4, C \cong C_3} Z_{G-V(C)}(t) < 0 \quad (t > 0).$$

The continuous-argument proof of Theorem ?? applies verbatim and gives $s^+(G) > s^-(G)$. Since $|E(G)| = n + 2$, this yields $s^+(G) > n + 2$.

Now suppose that the cyclic blocks are three triangles. If their vertex-disjoint packing number is at most two, Theorem ??(i) again gives $s^+(G) > s^-(G)$ and hence $s^+(G) > n + 2$.

It remains that the three triangle blocks are pairwise vertex-disjoint. Contract each triangle block to one marked vertex. Since every remaining block is a bridge, the resulting graph is a tree with three distinct marked vertices. Two edges of the minimal subtree joining the marks can be deleted so that the resulting three components contain one mark each. Each selected tree edge corresponds to an actual bridge of G . Deleting those two bridges therefore partitions $V(G)$ into three sets inducing connected graphs G_1, G_2, G_3 , each with exactly one triangle and otherwise only bridge blocks. Thus each G_j is a triangular unicyclic graph. Ning and Zeng [?] give

$$s^+(G_j) > |V(G_j)|.$$

Applying (??) to this induced vertex partition gives

$$s^+(G) \geq \sum_{j=1}^3 s^+(G_j) > \sum_{j=1}^3 |V(G_j)| = n.$$

This completes all cases. □

We next state a convenient cactus family. A triangular cactus is a cactus whose cyclic blocks are triangles. A bouquet of triangles is a set of triangle blocks having a common vertex. We say that the triangles of a triangular cactus form at most two bouquets if its set of triangle blocks can be partitioned into at most two subfamilies, each subfamily having a common vertex. Empty subfamilies are permitted.

Corollary 4.3 (One and two bouquets). *Let G be a triangular cactus containing at least one triangle. If its triangles form at most two bouquets, then*

$$s^+(G) > s^-(G).$$

This includes friendship graphs and double friendship graphs, with arbitrary trees attached at their vertices and with arbitrary forest components added.

Proof. All cycles are triangle blocks. A pairwise vertex-disjoint collection contains at most one triangle from each bouquet, and hence has size at most two. Apply Theorem ??(i). Tree attachments and forest components create no cycles and do not affect this verification. □

Here a friendship graph means a bouquet of triangles with one common center. A double friendship graph means any triangular cactus whose triangle blocks are the union of two specified bouquets; the two centers may be joined by a path or may coincide. This definition states exactly the property used in Corollary ??, rather than relying on a stronger triangle-intersection classification.

The phase-reversed version is equally immediate.

Corollary 4.4. *Let G contain a cycle, and suppose every cyclic block is a cycle of length $1 \pmod{4}$. If these cycle blocks can be partitioned into at most two bouquets (each subfamily has a common vertex), then $s^+(G) < s^-(G)$.*

Proof. Every cycle is a cyclic block and has length $1 \pmod{4}$, while at most one cycle from each bouquet can occur in a vertex-disjoint collection. Thus $\nu_o(G) \leq 2$, and Theorem ??(ii) applies. □

Remark 4.5 (Why the packing threshold appears). For three vertex-disjoint cycles of the common odd phase, the three-cycle terms in (??) are imaginary and have the opposite sign from the one-cycle terms: $(-2i)^3 = 8i$ in the $3 \pmod{4}$ case and $(2i)^3 = -8i$ in the $1 \pmod{4}$ case. Positivity of the signless matching factors alone therefore no longer fixes the imaginary sign. The argument proves the packing-two theorem sharply as a phase-counting mechanism; it does not assert that every graph with packing number three violates the conclusion.

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